Midterm Project

Kevin Lu

117926253

BAN 100 – Statistic for Analytics

Samaneh Gholami

October 22nd, 2025

Tables of Contents

Problem 1: Stock Ownership and Age Groups – Page 3

Problem 2: Number of Jobs, Gender, and Education – Page 9

Works Cited – Page 14

Problem 1: Stock Ownership and Age Groups

Analyst was interested in finding out whether the ownership of stocks varied by age. So, the research question would be "Do these data allow the analyst to find out if there are any differences in stock ownership between the four-age group? So, the goal is to determine if there are differences in stock vs four-age group.

Age Categories:

Young (under 35)

Early middle age (35 to 40)

Late middle age (50 to 65)

Senior (65+)

Hypotheses of One-Way ANOVA:

<u>Null hypothesis:</u> There is no difference in mean stock ownership among the four-age group:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

<u>Alternative hypothesis:</u> At least one of age categories differs with respect to mean stock ownership:

$$H_1: Not\ all\ the\ means\ are\ equal$$

Type of Test:

Therefore, I am going to conduct One-Way ANOVA because in this research question, it is more like evaluating the difference among the means of four-age categories so the assumptions could be samples can randomly and independently drawn. It can be done in SAS software.

Conditions / Assumptions Check:

First, as shown in figure 1.1, "Finding the Means of Age Categories", the mean and the standard deviation of the four-age category variables which shows Young with the mean of 44.40 and the standard deviation is 19.66 with 84 observations. For Early middle age, the mean is 52.47 and the standard deviation is 21. 67 with 131 observations. Late middle age average is 51.14 and the standard deviation is 21.72 with 93 observations. Lastly, for Senior, the average is 51.84 and the standard deviation is 21.09 with 58 observations.

Finding The Means of Age Categories

The MEANS Procedure

Variable	Label	N	Mean	Std Dev
Young	Young	84	44.40	19.66
Early_Middle_Age	Early_Middle_Age	131	52.47	21.67
Late_Middle_Age	Late_Middle_Age	93	51.14	21.72
Senior	Senior	58	51.84	21.09

Figure 1.1: Tables of "Finding the Means of Age Categories"

Since, the dataset as provided is wider. To run One-Way ANOVA test, it needs to have long dataset. So, I adjusted the dataset by converting from wide to long dataset. So therefore, I add a categorical variable called "ID" as this would be easier to proc transpose with four-age categories sort by "ID" variables. Then, after I convert from wide to long data, I rename the "COL1" to "Proportion" to make this dataset more sense. As shown in the figure 1.2 "Updated – Long Datasets" where it shows an updated datasets with the first 10 observations.

Updated - Long Datasets

Obs	ID	AgeGroup	_LABEL_	Proportion
1	1	Young	Young	24.8
2	1	Early_Middle_Age	Early_Middle_Age	28.9
3	1	Late_Middle_Age	Late_Middle_Age	81.5
4	1	Senior	Senior	66.8
5	2	Young	Young	35.5
6	2	Early_Middle_Age	Early_Middle_Age	7.3
7	2	Late_Middle_Age	Late_Middle_Age	0.0
8	2	Senior	Senior	77.4
9	3	Young	Young	68.7
10	3	Early_Middle_Age	Early_Middle_Age	61.8

Figure 1.2: Table of Updated – Long Datasets in the first 10 Observations

Box Plot:

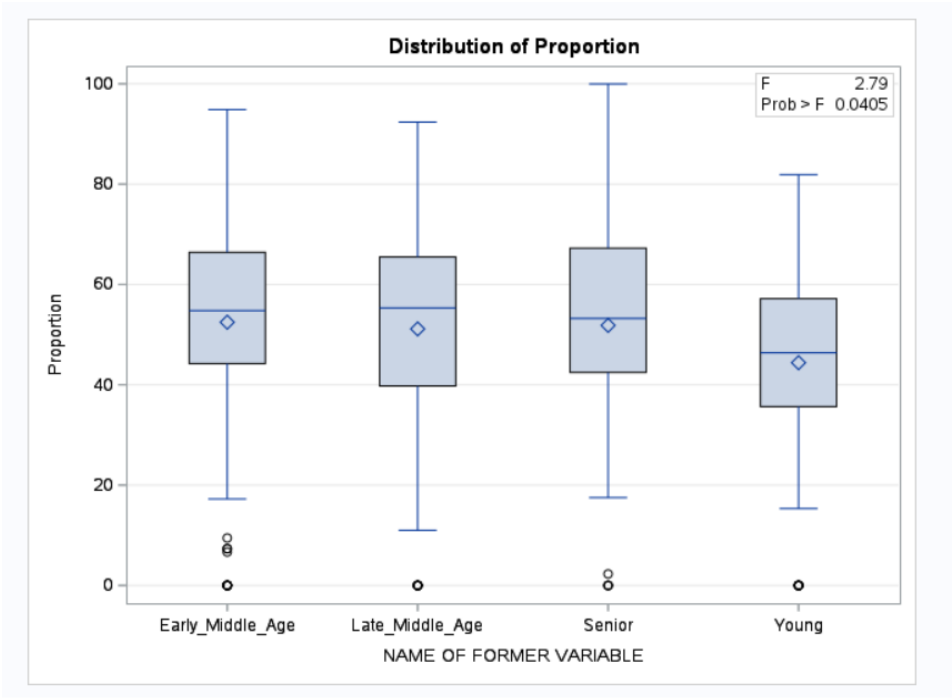

Figure 1.3: Box Plot of Age Group vs Proportion

Based on the result of Box Plot as shown in figure 1.3 "Box Plot of Age Group vs Proportion" it shows Early middle age, Late middle age, and senior groups categories are showing similar median values like roughly 50-55. Young age group is the one with the lowest median values with estimated to be 47 to 50. The ranges of Proportion is between 0 and 100. For the means as shown diamond markers, Early middle age, Senior, and Young are close to medians which could be symmetric distributions. Therefore, it is difficult to evaluate the normality from the boxplot so lets move on to Q-Q plot to evaluate the assumptions of normality where this will take a closer look at normality.

Q-Q Plot

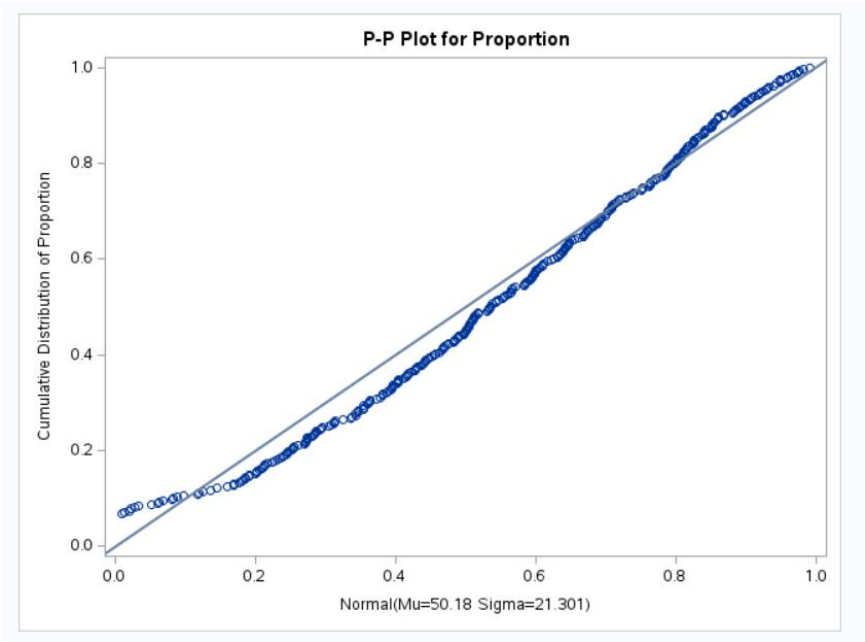

Figure 1.4: Q-Q Plot for Proportion

From figure 1.4 "Q-Q Plot for Proportion", it evaluates the assumption of normality as shown on the results, it is showing the dispersion of the data points along diagonal line in the plot, it considered as normality as this can tell. However, it can see that data points are located close to the diagonal line as this results, it is showing normally distributed. The distributed parameters are MU = 50.18 and Sigma = 21.301, it shows good fit. Therefore, it supports normality assumption as it is satisfied for an One-Way ANOVA Test.

Test Results:

Result of ANOVA Test

The ANOVA Procedure

Class Level Information		
Class	Levels	Values
AgeGroup	4	Early_Middle_Age Late_Middle_Age Senior Young

Number of Observations Read	544
Number of Observations Used	366

Result of ANOVA Test

The ANOVA Procedure

Dependent Variable: Proportion

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3741.3636	1247.1212	2.79	0.0405
Error	362	161870.9817	447.1574		
Corrected Total	365	165612.3453			

R-Square	Coeff Var	Root MSE	Proportion Mean
0.022591	42.14046	21.14610	50.18003

Source	DF	Anova SS	Mean Square	F Value	Pr > F
AgeGroup	3	3741.363610	1247.121203	2.79	0.0405

Figure 1.5: Result of ANOVA Test

Model DF = K-1 Error DF = N – (K – 1) Corrected Total DF = N - 1

Model DF = 4-1 Error DF = 365 – 4 Corrected Total DF = 366 – 1

Model DF = 3 Error DF = 362 Corrected Total DF = 365

F = MSA / MSW

F = 1247.1212 / 447.1574

F = 2.79

P-value = 0.0405

Decision and Interpretation

Significance value = 0.05

P-value = 0.0405

P-value < Significance value (0.0405 < 0.05)

Since p-value is less than significance value (0.0405 < 0.05), we can reject the null hypothesis, therefore we can conclude that at least one of age categories differs with respect to mean stock ownership.

Yes, based on the One-Way ANOVA test, these data allow the analyst to determine that there are differences in stock ownership between the four-age group.

Problem 2: Number of Jobs, Gender, and Education

In this research, measure of the health of national economy is concerns because of rapidly creating jobs. The issues is the number of jobs individuals hold. So, survey conducted where the aged between 17 and 45 were asked how many jobs they did work during their lifetimes. It was recorded with gender and gender attainment. The education categories are:

E1 = Less than high school

E2 = High school

E3 = Some college/university but not degree

E4 = At least one university

Type of Test:

Therefore, I am going to conduct Two-Way ANOVA because in this research question, it more like in each sample is defined in two ways and can put into two categorical group. It going to examines if the two factors affect each other to influence the continuous variable. It compares multiple groups of two factors. Since, it allows to compare between the means of three or more groups of data where two independent variables are being considered. It can be done in SAS software.

Conditions / Assumptions Check:

Since Two-Way ANOVA only works on long datasets and since datasets as provided, it is wide datasets as seen in figure 2.1 "Original Datasets" as SAS output for the first five observations so the ANOVA test may not work on wide datasets. To convert from wide datasets to long datasets, first it added a categorical variable called ID, using array data structure as it helps to organized which one belongs to Male while other belongs to Female. Like grouping by Gender. Based on separate gender observation, I use do statements whether which observation belongs to either "Male" or "Female". After I convert from wide to long datasets, I drop Male_E1 to Male_E4 and Female_E1 to Female_E4 as these are not needed to examine it. As shown in figure 2.2 "Updated Datasets" with he first ten observations.

Original Datasets

Obs	Male_E1	Male_E2	Male_E3	Male_E4	Female_E1	Female_E2	Female_E3	Female_E4
1	10	12	15	8	7	7	5	7
2	9	11	8	9	13	12	13	9
3	12	9	7	5	14	6	12	3
4	16	14	7	11	6	15	3	7
5	14	12	7	13	11	10	13	9

Figure 2.1: Original Datasets for the first Five Observations

Updated datasets

Obs	ID	i	Gender	Education	NumJobs
1	H1	1	Male	E1	10
2	H1	2	Male	E2	12
3	H1	3	Male	E3	15
4	H1	4	Male	E4	8
5	H1	1	Fema	E1	7
6	H1	2	Fema	E2	7
7	H1	3	Fema	E3	5
8	H1	4	Fema	E4	7
9	H2	1	Male	E1	9
10	H2	2	Male	E2	11

Figure 2.2: Updated datasets for the first Ten Observations

Box plot

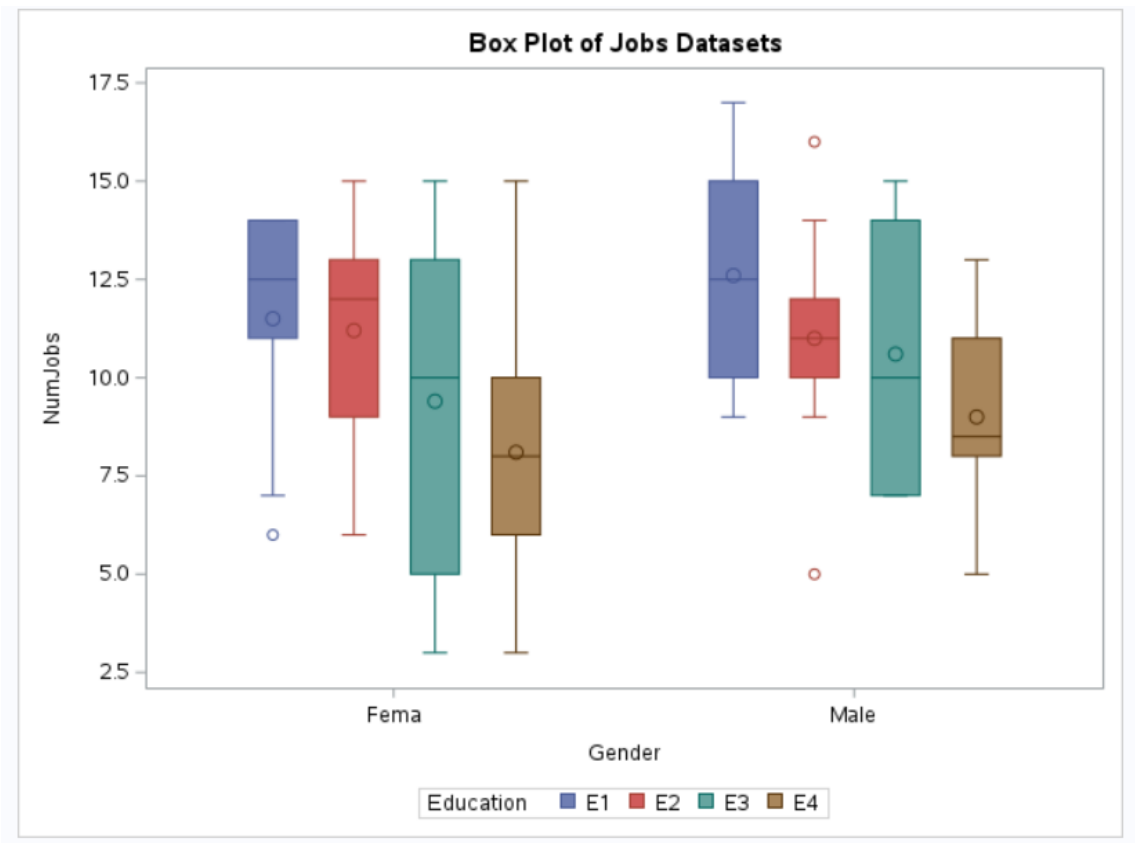

Figure 2.3: Box Plot of Jobs Datasets

In figure 2.3 “Box Plot of Jobs Datasets” the color of box based on Education, blue is for “Less than high school (E1)”, red is for “High school (E2)”, teel is for “Some college/university but not degree (E3)”, and brown is for “At least one university (E4)”. In the Gender, Female is on the left-hand side while the Male is on the right-hand side. As the finding, the Male E1 (blue on the right-hand side) shows the highest median number of jobs with roughly 12.5. Female E4 (brown on the left-hand side) shows the lowest median with roughly 7 or 8. On the right-hand side, male groups have higher job numbers than female groups across education levels.

Interaction Plot

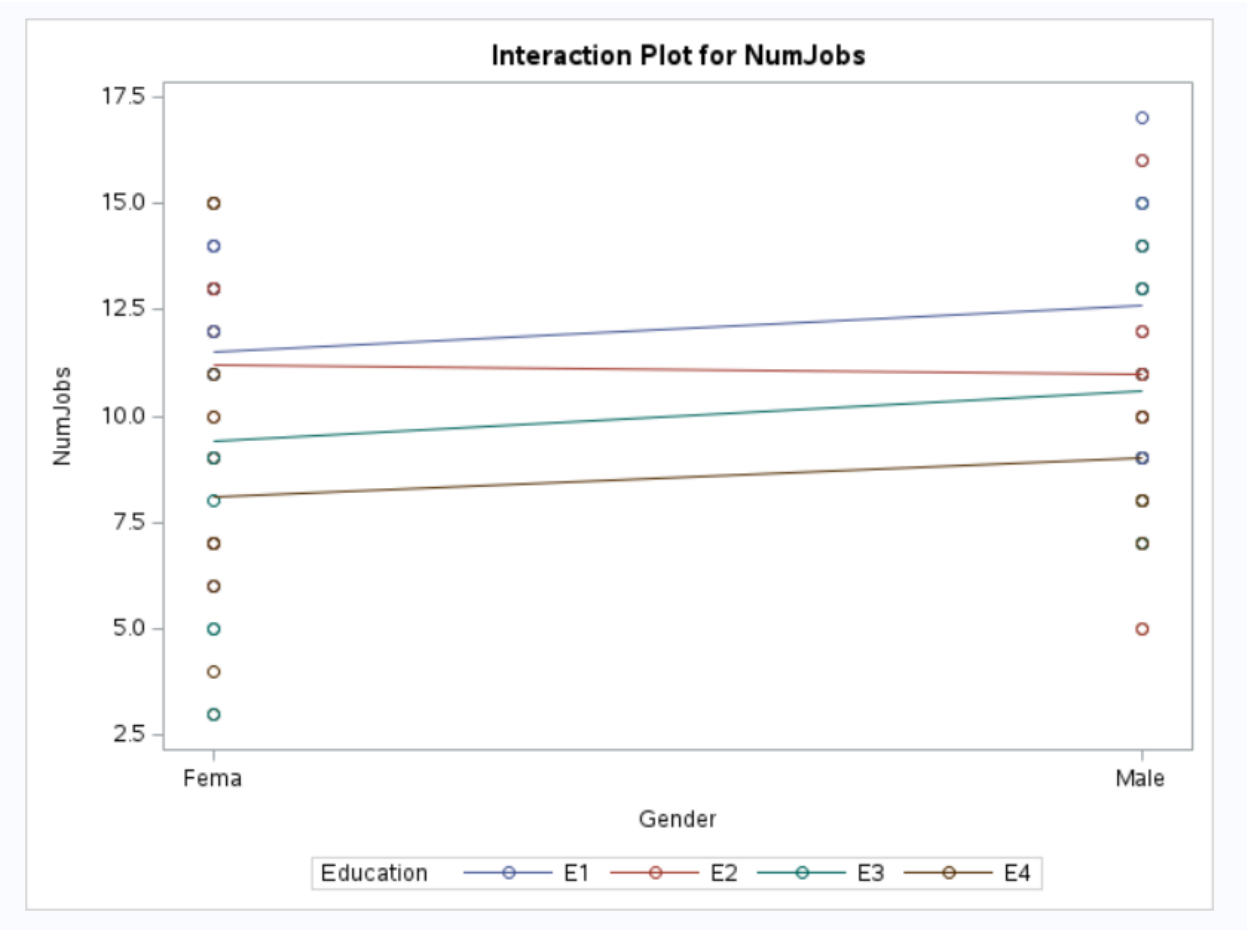

Figure 2.4: Interaction Plot for Number of Jobs

In figure 2.4 “Interaction Plot for Number of Jobs” as the result findings, all lines are all parallel which means all are parallel lines. This means, parallel lines are considered as there is no interaction that are present.

Hypothesis of Two-Way ANOVA:

A. H_o: *There is no interaction between gender and education in holding jobs*

H_1: *There is interaction between gender and education in holding jobs*

B. H_o: *The means of the men and women are equal*

H_o: *The means of the men and women groups are different*

C. H_o: *The means of holding jobs between the educational levels are equal*

H_o: *The means of holding jobs between the educational levels are different*

Test and Summary of Findings Section:

Class Level Information		
Class	Levels	Values
Gender	2	Fema Male
Education	4	E1 E2 E3 E4

Number of Observations Read	320
Number of Observations Used	80

Results of Two-Way ANOVA Test on jobs datasets

The GLM Procedure

Dependent Variable: NumJobs

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	153.3500000	21.9071429	2.17	0.0467
Error	72	726.2000000	10.0861111		
Corrected Total	79	879.5500000			

R-Square	Coeff Var	Root MSE	NumJobs Mean
0.174351	30.46392	3.175864	10.42500

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Gender	1	11.2500000	11.2500000	1.12	0.2944
Education	3	135.8500000	45.2833333	4.49	0.0060
Gender*Education	3	6.2500000	2.0833333	0.21	0.8915

Figure 2.5: Results of Two-Way ANOVA Test on jobs datasets

A. p = 0.8915 for Gender*Education: Since p-value = 0.8915, we cannot reject the null hypothesis and therefore conclude that there is no interaction between gender and education in holding jobs.

B. p = 0.2944 for Gender: Since p-value = 0.2944, we cannot reject the null hypothesis and therefore conclude that there are no differences in holding jobs between men and women.

C. p = 0.0060 for Education: Since p-value = 0.0060, we reject the null hypothesis and therefore conclude that there are differences in holding jobs between the educational levels.

Therefore, the issue has been resolved as there are no interaction between gender and education in holding jobs, no differences in holding jobs between men and women are these are considered equal and lastly, there are differences in holding jobs between the educational levels.

Works Cited

Gholami, Samaneh. “One-Way Analysis of Variance.” BAN100NAA. Seneca Polytechnic, 2025, North York, Ontario. BAN100NAA - One-Way Analysis of Variance Lecture Notes

Gholami, Samaneh. “Two-Way ANOVA.” BAN100NAA. Seneca Polytechnic, 2025, North York, Ontario. Two-way ANOVA Lecture notes - BAN10NAA